Integrated technology to produce bioethanol from whole-plant corn and life cycle assessment

Author links open overlay panelZhongfeng Geng [a b c], Zitian Guo [a b c], Shuyuan Yang [a b c], Huisheng Lyu [a b c], He Dong [a b c]

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Highlights

- An integrated technology for bioethanol production from whole maize (including C5 and C6 sugars) is newly proposed.

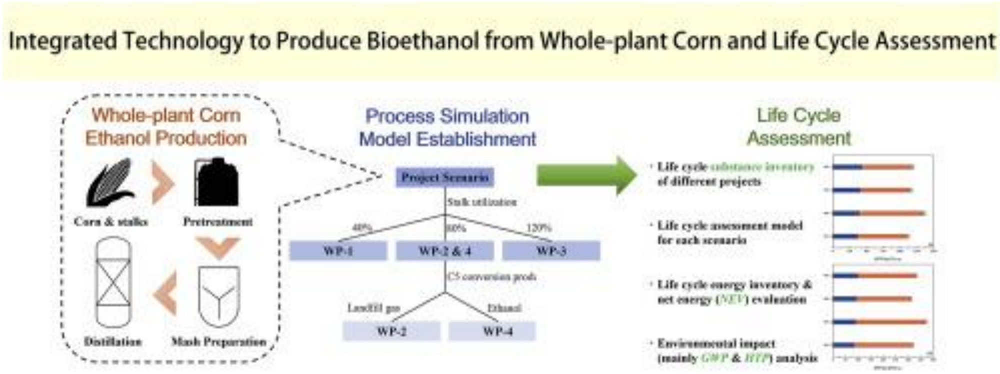
- The ratio of utilized straw to corn of 80 % in WP-2 was the optimal whole-plant corn ethanol feedstock proportioning scheme.

- Compared to gasoline products with the same energy content, the WP-4 scenario has lower E_{inf}, *NEV* and *GWP* ratings.

Abstract

An integrated bioethanol production technology using whole-plant corn including C5 sugars and C6 sugars was proposed. Life cycle assessment was employed to investigate the effects of different utilization modes of C5 sugars on bioethanol production. The results showed that the whole-plant corn ethanol WP-4 process based on 80 % straw utilization and C5 sugar conversion for ethanol utilization was a superior scheme with substantial application and promotion potential. Compared with gasoline products of the same energy content, the E_{inf}, *NEV* (net energy value) and *GWP* (global warming potential) values were reduced by 73.8 %, 38.2 % and 59.6 %, respectively. The WP-4 process exhibited higher energy yield, greater environmental benefits and more remarkable carbon neutralization potential.

Graphical abstract



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Introduction

Bioethanol is an important form of renewable energy that can be used as a liquid fuel for automobiles, which differentiates it from other renewable energy sources like wind and solar [1]. As bioethanol was produced from biomass, this liquid fuel can reduce carbon emission, improve the ecological environment and promote the development of agriculture [2]. The use of corn [3] and other grain raw materials to produce bioethanol [4] suffer from insufficient supply for its competition with food. Therefore, the use of cellulosic raw materials based on straw has attracted more and more attention [5,6].

Biorefinery aim to maximize the value of biomass by converting it into a variety of products such as biofuels, biochemicals and biomaterials [[7], [8], [9]]. Sriariyanun et al. [10] produced cellulosic ethanol through a biorefinery process using genetically engineered microorganisms. Corn straw produces not only ethanol but also furfural and lignin-derived carbon fiber.

There have been some previous studies on cellulosic bioethanol [1,4,7]. Lyu et al. [11] proposed a hydrothermal process to pretreat straw for cellulosic bioethanol production enhanced by by-product organic acids. Hydrothermal pretreatment has emerged as an environmentally favorable approach for lignocellulosic biomass conversion, using water at elevated temperatures (160–240 °C) and pressure to solubilize hemicellulose while making cellulose more accessible for enzymatic hydrolysis [[12], [13], [14]]. This pretreatment method is highly efficient, can avoid equipment corrosion and fermentation inhibition that exist in traditional acid-based and base-based pretreatment process [[15], [16], [17]]. However, this process still calls for an improvement in the efficiency of enzyme fermentation and in the reduction of energy, material and water consumption [[18], [19], [20]].

This article aims to proposes a whole-plant corn fuel ethanol production technology based on autocatalytic hydrothermal treatment of cellulose raw materials [11] and on integration of cellulose enzymatic digestion and corn mash preparation. This technology would combine the advantage of corn ethanol and cellulosic ethanol [21], and might overcome the difficulties faced by the current technology and industrial production of bioethanol from cellulose raw materials such as corn straw. Current challenges in cellulosic bioethanol production include inefficient C5 sugar fermentation, high energy consumption in pretreatment, and competition with food resources [3,7]. This work addressed these issues by minimizing the use of chemicals through the use of autocatalytic hydrothermal pretreatment [11], co-fermentation of C5 and C6 sugars to increase yields, and optimizing energy recovery through combustion of lignin residues.

With regard to the evaluation of bioethanol technologies and projects, investment and cost of production are usually emphasized. However, environmental and human health impacts such as CO_2 and other greenhouse gas emissions [22] was just simply descripted in literature. There is a lack of detailed data for evaluation and analysis of carbon emissions in the whole life cycle of ethanol production [9], such as raw material types, cultivation, transportation, production, transportation and combustion use, which is required by national policy sometimes. Therefore, a comprehensive, scientific and fair evaluation on environmental impacts of existing and developing fuel ethanol projects during the life-cycle [8] will be beneficial to formulate low-carbon emission programs and to optimize strategy for fuel ethanol production.

Life Cycle Assessment (LCA) is a systematic approach evaluate the environmental impacts for all stages of a product or service's life, from the extraction of raw materials, production, use, and maintenance, to its final disposal or recycling. It helps to identify and quantify the environmental burdens and potential impacts, such as energy consumption, greenhouse gas emissions, water usage, and waste generation, enabling a comprehensive understanding of the environmental performance and aiding in decision-making to reduce environmental footprints [23,24]. The GaBi [25] life cycle assessment model was adopted to assess the life cycle carbon emissions and other environmental impact indicators of the proposed whole-plant corn fuel ethanol technology. A solution to the technological development dilemma faced by cellulosic ethanol might be provided.

In this paper, basic data such as material and energy accounting at the production stage required for the life cycle evaluation of the corresponding whole-plant corn fuel ethanol products were obtained. And relevant literature data at the stages of raw material cultivation and product transportation were collected and analyzed to establish the list of substances for the LCA. And then, the GaBi evaluation model was used to evaluate carbon emissions and other environmental impacts of the life cycle of this whole-plant corn fuel ethanol production. Further carbon emission reduction strategies for whole-plant corn fuel ethanol production were proposed.

Scenario settings

This study was conducted on a multi-feedstock bioethanol project with feedstock of corn and corn straw feedstock produced in Jilin Province, China. In this study, the whole-plant corn ethanol project will use process scenarios such as secondary cellulose pretreatment of corn straw, C5 sugar conversion to biogas or ethanol, C6 sugar [26] fermentation for ethanol and by-production of DDGS [27] products. As shown in Table 1, the study establishes WP-1, WP-2, WP-3 and WP-4 project scenarios based

Results and discussion

This study proposes an integrated bioethanol production technology utilizing whole corn plants, including C5 and C6 sugars, and evaluates its energy and environmental performance through a life cycle assessment (LCA). The results indicate that the WP-4 scheme demonstrates higher energy efficiency and environmental benefits compared to other schemes.

Conclusion

This paper proposed an integrated bioethanol production technology using whole-plant corn including C5 sugars and C6 sugars. Four whole-plant scenarios (WP-1, WP-2, WP-3 and WP-4) were set up to investigate the effects of different utilization modes of C5 sugars on bioethanol production. The data of material and energy balances under the relevant scenarios were obtained. And then, the GaBi evaluation model was employed to conduct a life cycle assessment of the whole-plant corn ethanol project.

CRediT authorship contribution statement

Zhongfeng Geng: Visualization, Validation, Supervision, Software. **Zitian Guo:** Methodology, Formal analysis, Data curation, Conceptualization. **Shuyuan Yang:** Validation, Supervision, Software, Investigation. **Huisheng Lyu:** Project administration, Methodology, Data curation, Conceptualization. **He Dong:** Supervision, Software, Resources.

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